

Program : Diploma in Textile Technology	
Course Code : 3065	Course Title: Yarn manufacture I Lab
Semester : 3	Credits: 1.5
Course Category: Program core	
Periods per week: 3 (L:0, T:0 P:3)	Periods per semester: 45

Course Objectives:

- To impart practical knowledge about the material passage in the spinning preparatory machines, concept of gearing diagram, parts identification and carry out production, draft and twist calculations

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I	1

Course Outcomes:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Understand the sequence of Blow room machines, gearing diagram and calculate draft, production and efficiency.	12	Applying
CO2	Illustrate the passage of material, gearing diagram, and settings of carding machine. Calculate its draft constant, change wheel, production and efficiency.	12	Applying
CO3	Illustrate the material passage, drafting roller setting, gearing diagram of draw frame and calculate draft constant, production and efficiency.	11	Applying
CO4	Understand the construction details and combing cycle of comber machine and carry out production and efficiency calculation	7	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	1					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive levels
CO1	Understand the sequence of Blow room machines, gearing diagram and calculate draft, production and efficiency.		
M1.01	Draw the line sketch of modern blow room line according to type of raw material	2	Understanding
M1.02	Sketch the material passage of Scutcher machine and identify parts.	2	Understanding
M1.03	Draw the gearing diagram of scutcher and determine draft and production of scutcher	4	Applying
M1.04	Study of method of determining cleaning efficiency of openers and blow room	2 2	Applying
M1.05	Determine the lap hank and beats/inch of bladed beater in scutcher	2	Applying
CO2	Illustrate the passage of material, gearing diagram, and settings of carding machine. Calculate its draft constant, change wheel, production and efficiency.		
M2.01	Study of Construction details and the material passage of carding machine and identify the parts	3	Understanding
M2.02	Draw the gearing diagram of card and determine the draft constant and draft change wheel for different hank of sliver.	4	Applying
M2.03	Study of wire point specifications and settings in carding machine as per the given mixing.	3	Understanding
M2.04	Determination of production and efficiency of carding machine	2	Applying
	Lab Exam – I	1.5	

CO3	Illustrate the material passage, drafting roller setting, gearing diagram of draw frame and calculate draft constant, production and efficiency.		
M3.01	Sketch the material passage of draw frame machine and identify the parts.	3	Understanding
M3.02	Draw the gearing diagram of draw frame and determine the draft constant and draft change wheel for different hank of sliver.	4	Applying
M3.03	Determination of the production and efficiency of draw frame machine	2	Applying
M3.04	Study of drafting roller setting and process parameters in draw frame.	2	Understanding
CO4	Understand the construction details and combing cycle of comber machine and carry out production and efficiency calculation.		
M4.01	Study of Construction details and the combing cycle of comber machine	4	Understanding
M4.02	Calculate the production and efficiency of comber machine	3	Applying
	Lab Exam - II	1.5	

The students should keep record for Yarn manufacture I Practical and submit certified & bonafide record at the time of practical examination.

Text / Reference:

T/R	Book Title/Author
T1	Laboratory manual
R2	R. Gokarneshan, Mechanics and calculations of Textile Machinery
R3	Oxtoby E., “Spun Yarn Technology”, Butterworths, London, 2002
R4	W.S. Taggart, Cotton spinning calculation
R5	T.K.Pattabhiram-Essential calculations of Practical Cotton spinning

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in
2	www.reiter.com
3	www.laksmimach.com

SCHEME OF EVALUATION OF END SEMESTER EXAMINATION

Sl.No	Criteria	Marks	Weightage
1	Design and procedure of the problem	15	30%
2	Implementation and output of the problem	20	40%
3	Viva Voce	10	20%
4	Record	5	10%
	TOTAL	50	